ROBOTIC ARM SIMULATION WITH PICK-AND-PLACE OPERATION

Task 2–Robotic ArmSimulation

Submitted to:
HorizonTechX Internship Program

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Internship Domain:
Robotics and Automation

Date:
20-06-2026

INTRODUCTION

Aroboticarm is aprogrammable mechanical device designed to perform tasks such as picking, placing, lifting, and manipulating objects with precision. Robotic arms are widely used in manufacturing industries, healthcare systems, warehouses, logistics, and research laboratories to improve efficiency, accuracy, and automation.

This project presents a robotic arm mounted on a mobile platform equipped with Mecanum wheels. The robotic arm consists of multiple joints and a gripper mechanism that enable it to perform basic pick-and-place operations. The mobile platform allows omnidirectional movement, making the system more flexible and suitable for automation applications.

The primary objective of this project is to demonstrate the working principles of robotic manipulation, Degrees of Freedom (DOF), motion control, and object handling through a simulated robotic arm model.

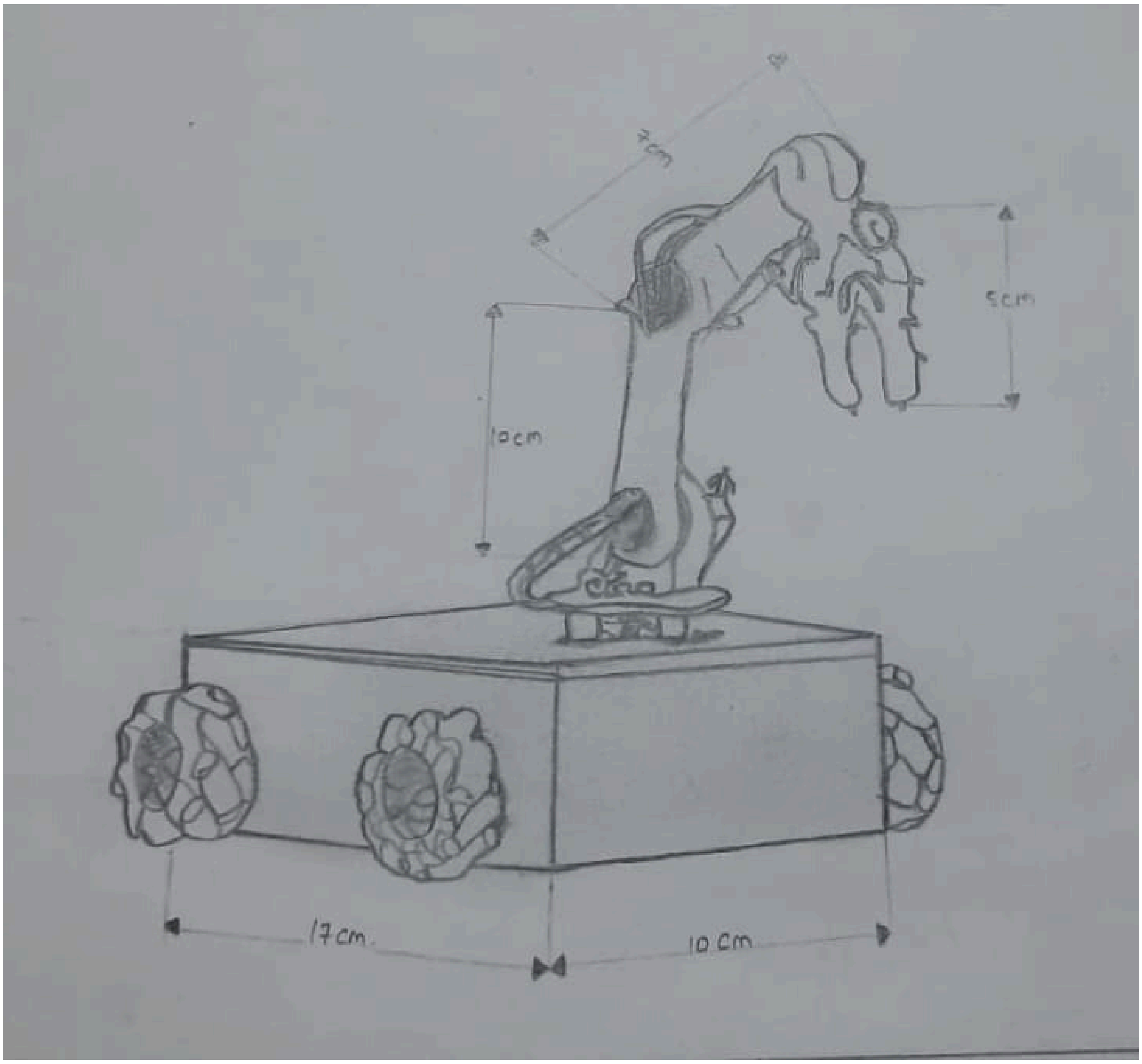


Figure 1: Initial Design Sketch of the Robotic Arm

PROJECT OBJECTIVES

The main objectives of this robotic arm simulation project are:

• To design and develop a robotic arm capable of performing basic pick-and-place operations.

• To understand the concept of Degrees of Freedom (DOF) in robotic systems.

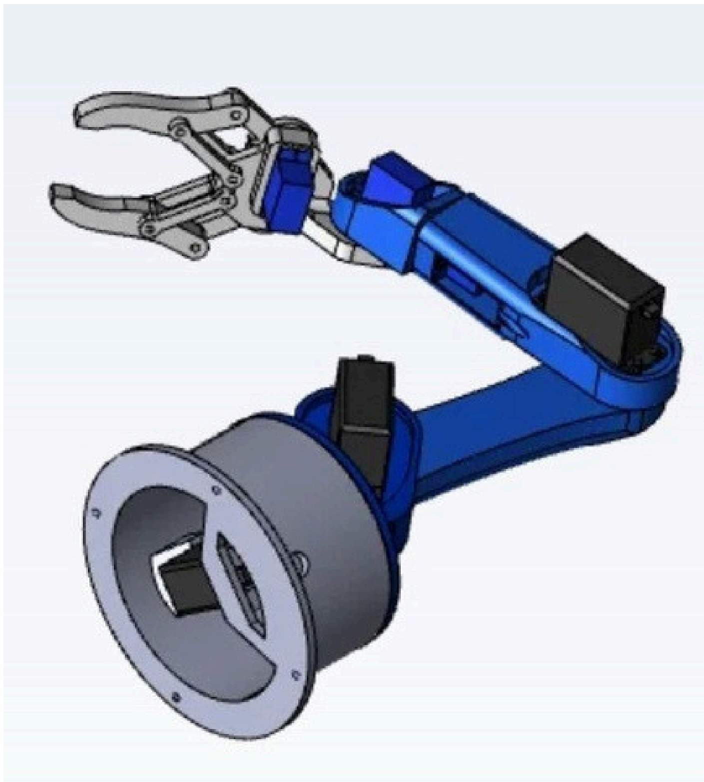
• To simulate the movement of robotic joints and gripper mechanisms.

• To study robotic manipulation and object handling techniques.

• To demonstrate the integration of a robotic arm with a mobile platform.

• To gain practical knowledge of robotics, automation, and motion control systems.

• To explore the applications of robotic arms in industrial and service environments.



MECHANICAL DESIGN OF THE ROBOTIC ARM

Therobotic systemconsists of a mobile platform integrated witha robotic arm and gripper mechanism. The mobile base is equipped with four Mecanum wheels, enabling omnidirectional movement in different directions. The robotic arm is mounted on the platform and consists of multiple joints that provide flexibility and precise movement.

The major components of the system are:

1. Mobile Chassis – Provides structural support for all components.
2. Mecanum Wheels – Enable multidirectional movement of the platform.
3. Base Joint – Allows rotation of the robotic arm.
4. Shoulder Joint – Controls vertical movement of the arm.
5. Elbow Joint – Extends and retracts the arm.
6. Wrist Joint – Provides orientation control.
7. Gripper Mechanism – Grasps and releases objects during pick-and-place operations.

The combination of these components allows the robotic arm to perform object manipulation tasks efficiently while maintaining mobility through the mobile platform.

Table

Component	Function
Mobile Chassis	Supports the robotic system
Mecanum Wheels	Omnidirectional movement
Base Joint	Rotational movement
Shoulder Joint	Vertical arm movement
Elbow Joint	Arm extension
Wrist Joint	Orientation control
Gripper	Object handling

Figure 3: Mechanical Structure of the Robotic Arm System

DEGREES OF FREEDOM (DOF)

Degrees of Freedom (DOF) refer to the number of independent movements that a robotic system can perform. Each joint in a robotic arm contributes to its overall flexibility and range of motion.

The robotic arm designed in this project consists of five Degrees of Freedom (5-DOF), enabling the arm to perform various movements required for pick-and-place operations.

The individual Degrees of Freedom are:

1. Base Rotation – Allows the arm to rotate horizontally.
2. Shoulder Joint – Moves the arm upward and downward.
3. Elbow Joint – Extends and retracts the arm.
4. Wrist Joint – Adjusts the orientation of the end-effector.
5. Gripper Mechanism – Opens and closes to grasp objects.

The combination of these five movements enables the robotic arm to accurately reach, pick, move, and place objects at desired locations.

Table

Joint	Movement	DOF
Base Joint	Horizontal Rotation	1
Shoulder Joint	Up/Down Movement	1
Elbow Joint	Extension/Retraction	1
Wrist Joint	Orientation Control	1
Gripper	Open/Close Action	1
Total		**5 DOF**

PICK-AND-PLACE SIMULATION

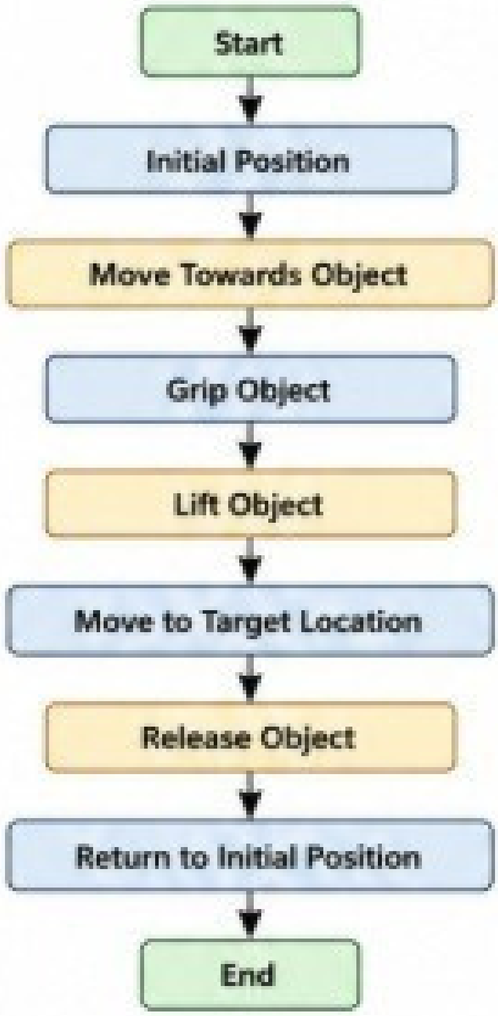
Pick-and-place is one of the most common operations performed by robotic arms in industrial automation. In this process, the robotic arm identifies an object, moves toward it, grasps it using the gripper, transports it to a target location, and releases it.

The robotic arm in this project performs the pick-and-place operation through coordinated movements of its joints and gripper mechanism. The operation is executed in a sequence of predefined steps to ensure accurate object handling.

Simulation Steps:

1. Initial Position – The robotic arm remains in its default position.
2. Object Detection – The arm identifies the object's location.
3. Arm Movement – The arm moves toward the object.
4. Gripping Action – The gripper closes and grasps the object.
5. Object Transportation – The arm carries the object to the target position.
6. Release Action – The gripper opens and releases the object.
7. Return Position – The arm returns to its initial position.

This simulation demonstrates the fundamental principles of robotic manipulation and automated material handling systems.



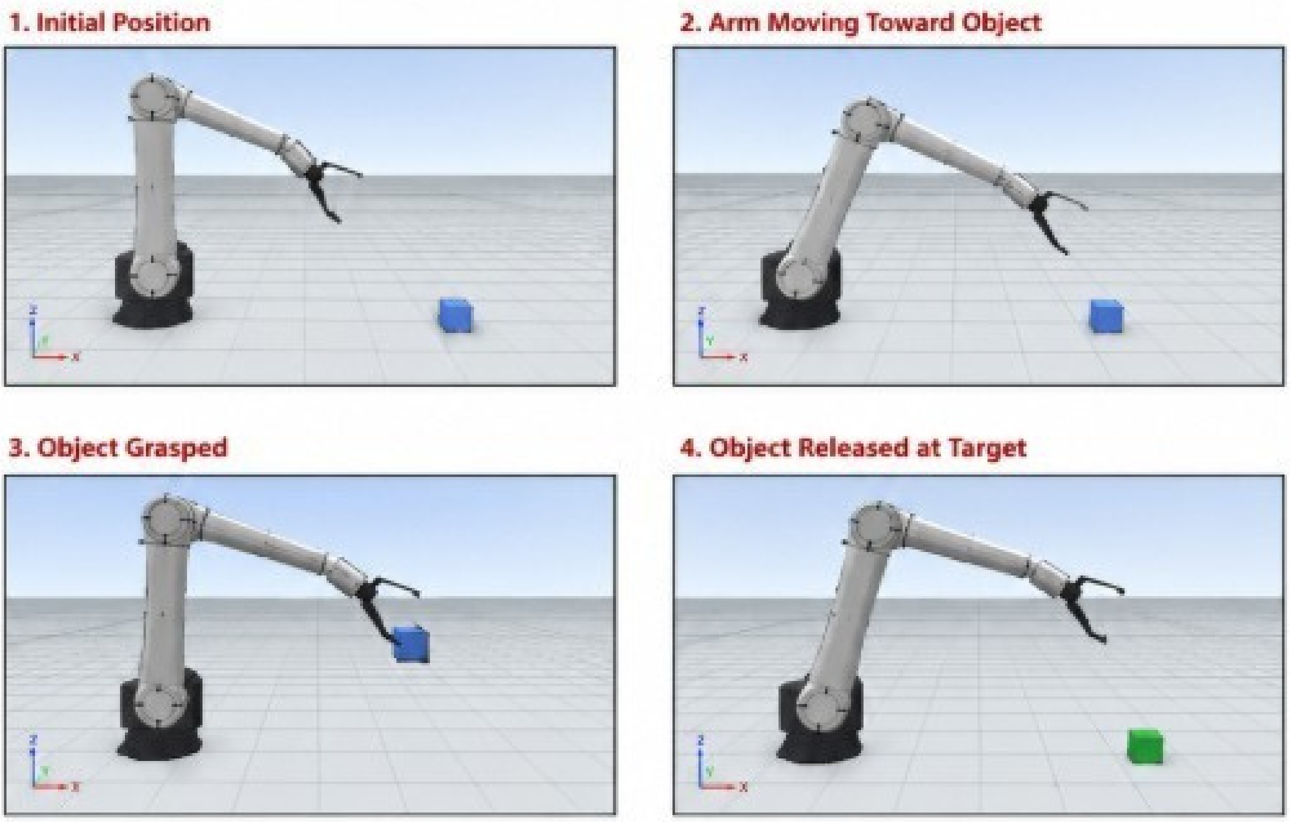


Figure 5: Pick-and-Place Operation Sequence

SIMULATION RESULTS AND OBSERVATIONS

The robotic arm simulation was successfully designed and tested using a CAD/simulation environment. The arm completed the pick-and-place operation by moving an object from the source location to the target location through coordinated joint movements.

Results Obtained:

- Successful movement of all joints according to predefined angles.
- Smooth pick-and-place operation without collision.
- Accurate positioning of the gripper at the target location.
- Proper object handling during gripping and releasing stages.
- Stable operation throughout the simulation cycle.

Observations:

1. Increasing the number of degrees of freedom improves flexibility and reachability.
2. Joint angle limits affect the workspace of the robotic arm.
3. Precise gripper control is essential for reliable object handling.
4. Simulation helps identify design issues before physical implementation.
5. Robotic arms can significantly improve automation efficiency in industrial applications.

The simulation validates the effectiveness of the designed robotic arm for performing simple material handling tasks.

Project Resources

CONCLUSION AND FUTURE SCOPE

Therobotic armmodel was successfully designedandsimulated using CAD/simulation software. The robotic arm consists of multiple joints providing the required degrees of freedom for performing pick-and-place operations. The simulation demonstrated smooth movement, object gripping, transportation, and placement at the desired target location.

Through this project, important robotics concepts such as joint movement, degrees of freedom, motion control, and workspace analysis were explored. The successful completion of the simulation shows how robotic arms can automate repetitive tasks with high precision and efficiency.

CODE IMPLEMENTATION

The robotic arm simulation was developed using JavaScript and Three.js to create an interactive 3D robotic system. The robotic arm consists of multiple joints, including the base, shoulder, elbow, wrist, and gripper. Each joint is programmed to perform specific movements required for pick-and-place operations.

The simulation controls the rotation and movement of individual joints to accurately reach an object, grasp it using the gripper, transport it to the target location, and release it. Real-time animation techniques were used to ensure smooth motion and realistic robotic behavior.

The project demonstrates the practical implementation of robotic arm kinematics, motion control, and object manipulation in a virtual environment. The complete source code for the project is available in the GitHub repository provided below.

Key Features:

• 3D robotic arm rendering using Three.js

• Joint-based movement control

• Pick-and-place simulation

• Real-time animation

• Interactive visualization

GitHub Repository:

https://github.com/bushrafaizi1162529-pixel/robotic-arm

Future Scope

• Integration of sensors for object detection and obstacle avoidance.

• Implementation of computer vision for intelligent object recognition.

• Addition of more degrees of freedom for increased flexibility.

• Development of autonomous motion planning algorithms.

• Integration with Industry 4.0 and IoT-based automation systems.

• Construction of a physical prototype based on the simulated model.

This project provides a foundation for understanding robotic manipulation and industrial automation technologies.

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[8] Arduino Documentation, https://docs.arduino.cc/ (if Arduino concepts were used)

[9] Robotics and Automation Educational Resources.

[10] Project source code and simulation files developed by the author.